

When does a directed container decompose?

The two-atoms conjecture and its freeness correction

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Abstract

A directed container is the same thing as a small category, and a recurring dream is that every small category should split cleanly into two pure “atoms”: a poset skeleton (the shape data, with no loops) and the endomorphism monoids sitting at each vertex (the pure loop data), glued together by composition as a mutual action — a Zappa–Szép product. I make this precise and then refute the naive form. The crisp result: for fixed objects a, b the source-oriented unique factorization $f = t \circ e$ (with $e \in \text{End}(a)$ an endomorphism at the source and t a transversal arrow) holds for every $f \in \text{Hom}(a, b)$ *if and only if* $\text{Hom}(a, b)$ is a *free* right $\text{End}(a)$ -set; dually for the target orientation and free left $\text{End}(b)$ -sets. A quick necessary numerical test in the finite case is that $|\text{End}(a)|$ divides $|\text{Hom}(a, b)|$. The smallest counterexample has five morphisms and fails this divisibility, so the Zappa–Szép decomposition does not exist in general. The correct general statement is a bimodule/profunctor assembly: each $\text{Hom}(a, b)$ is an $(\text{End}(b), \text{End}(a))$ -biset, and $\mathcal{C}$ is the collage glueing of its vertex endomorphism monoids along these connecting bisets. Zappa–Szép is exactly the free, one-sided special case. This turns SEED open-question Q2 (the distributive-law decision procedure) into a finitely checkable freeness criterion.

1 Setup: directed containers as small categories

I take for granted the equivalence between directed containers and small categories, and translate the dictionary once so that everything below is phrased categorically.

Definition 1 (Dictionary). A small category $\mathcal{C}$ corresponds to a directed container as follows: the *shapes* are the objects; the positions $P(s)$ over a shape s are the morphisms out of s ; the root position o is the identity id_s ; the operation $s \downarrow p$ takes the codomain of p ; and the position monoid operation $p \oplus q$ is composition *with order reversed*, $p \oplus q = q \circ p$. For objects a, b I write $\text{Hom}(a, b)$ for the set of morphisms $a \rightarrow b$, and

$$\text{End}(a) := \text{Hom}(a, a)$$

for the *endomorphism monoid* at a , a monoid under composition with unit id_a .

The two natural ways an endomorphism monoid acts on a hom-set are the whole content of what follows, so I state them carefully.

Definition 2 (The canonical biset structure). Fix objects a, b . The monoid $\text{End}(a)$ acts on the *right* of $\text{Hom}(a, b)$ by precomposition,

$$f \cdot e := f \circ e \quad (f \in \text{Hom}(a, b), e \in \text{End}(a)),$$

and the monoid $\text{End}(b)$ acts on the *left* by postcomposition,

$$e' \cdot f := e' \circ f \quad (e' \in \text{End}(b), f \in \text{Hom}(a, b)).$$

These two actions commute by associativity, $(e' \circ f) \circ e = e' \circ (f \circ e)$, so $\text{Hom}(a, b)$ is canonically an $(\text{End}(b), \text{End}(a))$ -*biset*. The right action is unital and associative because $\text{End}(a)$ is a monoid; likewise the left.

2 The two atoms and the naive conjecture

There are two “pure” kinds of small category, and the dream is that every category is built from just these.

Definition 3 (The two atoms).

1. A *poset* (more precisely a thin category): at most one arrow between any ordered pair of objects, so in particular every endomorphism monoid is trivial, $\text{End}(a) = \{\text{id}_a\}$. This is the pure *shape* atom — all skeleton, no loops.
2. A *monoid*: a one-object category, where every arrow is a loop. This is the pure *loop* atom — all endomorphism, no shape.

Conjecture 4 (Two atoms; SEED Q2, naive form). Every small category $\mathcal{C}$ is a Zappa–Szép product of a poset skeleton and its vertex endomorphism monoids, glued by composition as the mutual action. Concretely (*source orientation*): there is a sub-collection of *transversal* arrows, forming a thin subcategory, such that every $f \in \text{Hom}(a, b)$ factors *uniquely* as

$$f = t \circ e, \quad e \in \text{End}(a), t \text{ transversal.}$$

Dually (*target orientation*) every f factors uniquely as $f = e' \circ t$ with $e' \in \text{End}(b)$.

The conjecture bundles three claims: (a) such a factorization exists; (b) it is unique; (c) the transversal is canonically the poset/reachability skeleton. I will show all three can fail, pin the exact condition for (a)+(b), and exhibit a separate failure of (c).

3 The freeness correction

Recall that a right M -set X (for a monoid M) is *free on a basis* $T \subseteq X$ if the action map

$$\mu : T \times M \longrightarrow X, \quad \mu(t, m) = t \cdot m$$

is a bijection; equivalently $X \cong \coprod_{t \in T} M$ as right M -sets, each summand the regular action. X is *free* if it admits some basis. For a *group* M this is the same as the orbit-theoretic notion (every stabilizer trivial, with T one representative per orbit), but for a general monoid the action-map formulation is the correct definition. A *transversal* is precisely a basis: one representative whose translates $t \cdot M$ cover X disjointly and faithfully.

Theorem 5 (Freeness criterion). *Fix objects a, b of a small category $\mathcal{C}$. The source-oriented unique factorization of Conjecture 4 — every $f \in \text{Hom}(a, b)$ factors uniquely as $f = t \circ e$ with $e \in \text{End}(a)$ and t ranging over a fixed transversal — exists if and only if $\text{Hom}(a, b)$ is a free right $\text{End}(a)$ -set. In that case a transversal is exactly a choice of one representative arrow per $\text{End}(a)$ -orbit. Dually, the target-oriented factorization $f = e' \circ t$ with $e' \in \text{End}(b)$ exists iff $\text{Hom}(a, b)$ is a free left $\text{End}(b)$ -set.*

Proof. I prove the source statement; the target statement is the mirror image under $f \mapsto f^{\text{op}}$.

Unfold the factorization. A transversal $T \subseteq \text{Hom}(a, b)$ together with the right $\text{End}(a)$ -action gives the action map

$$\mu : T \times \text{End}(a) \longrightarrow \text{Hom}(a, b), \quad \mu(t, e) = t \circ e = t \cdot e.$$

The factorization claim “every f is $t \circ e$ for some transversal t and some $e \in \text{End}(a)$ ” is exactly surjectivity of μ ; “uniquely” is injectivity of μ . Hence: *source-oriented unique factorization at (a, b) holds $\iff$ there exists a transversal T for which μ is a bijection.*

But “ μ is a bijection for some T ” is, verbatim, the definition of $\text{Hom}(a, b)$ being a free right $\text{End}(a)$ -set with basis T . So the two statements are literally the same statement, and the theorem holds. I record the two directions explicitly for the reader.

($\Leftarrow$) If $\text{Hom}(a, b)$ is free with basis T , then μ is a bijection by definition; reading off $f = \mu(t, e) = t \cdot e$ gives existence (surjectivity) and uniqueness (injectivity).

($\Rightarrow$) Conversely, if some transversal T makes μ a bijection, then $\text{Hom}(a, b) \cong \coprod_{t \in T} \text{End}(a)$ as right $\text{End}(a)$ -sets via μ , which is exactly freeness with basis T .

The phrase “a transversal is a choice of one representative per orbit” is the group intuition; for a general monoid the precise statement is that a basis T indexes the summands of the free decomposition, and surjectivity of μ says T meets every right $\text{End}(a)$ -orbit while injectivity says each restricted orbit map $e \mapsto t \cdot e$ is injective *and* distinct basis elements have disjoint orbits. $\square$

Remark 6. The slickest phrasing: a transversal making μ bijective is the same datum as a free-basis of $\text{Hom}(a, b)$ as a right $\text{End}(a)$ -set, and a set has a basis as a free M -set iff the action is already free. “Existence of factorization” is the transversal meeting every orbit; “uniqueness” is freeness (no two endomorphisms send a transversal arrow to the same morphism).

Corollary 7 (Numerical necessary test). *If $\mathcal{C}$ is finite and the source-oriented unique factorization holds at a, b , then $|\text{End}(a)|$ divides $|\text{Hom}(a, b)|$. Dually $|\text{End}(b)|$ divides $|\text{Hom}(a, b)|$ for the target orientation.*

Proof. A free right $\text{End}(a)$ -set is a disjoint union of orbits each of size $|\text{End}(a)|$, so $|\text{Hom}(a, b)| = (\text{number of orbits}) \cdot |\text{End}(a)|$. $\square$

4 The smallest counterexample

Example 8 (Five morphisms, divisibility fails both ways). Let $\mathcal{C}$ have two objects a, b . Put $\text{Hom}(a, b) = \{s\}$ a single connecting arrow, $\text{Hom}(b, a) = \emptyset$, and

$$\text{End}(a) = \text{End}(b) = \mathbb{Z}/2 = \{1, g\}, \quad g^2 = 1,$$

with the loops fixing the connecting arrow on both sides:

$$g \circ s = s = s \circ g.$$

There are five morphisms in all: $\text{id}_a = 1_a$, g_a , $\text{id}_b = 1_b$, g_b , and s .

Proposition 9. *Example 8 is a genuine small category, and it violates the source- and target-oriented unique factorization simultaneously.*

Proof. It is a category. The only composites to define beyond the two copies of $\mathbb{Z}/2$ are those involving s . Since s is the unique arrow $a \rightarrow b$ and there are no arrows $b \rightarrow a$, every legal composite with s lands in $\{s\}$: $g_b \circ s$, $s \circ g_a$, $g_b \circ s \circ g_a$ are all forced to be s , which is exactly the prescription $g \circ s = s = s \circ g$ (the loops are “central” on s and fix it). Associativity holds because each $\mathbb{Z}/2$ is associative and any bracketing of a string containing s evaluates to s (every loop fixes s), independent of the bracketing; unit laws are immediate.

It fails unique factorization. Source orientation: $\text{End}(a) = \{1, g\}$ has order 2 but $|\text{Hom}(a, b)| = 1$, and $2 \nmid 1$, so by Corollary 7 source-UF fails. Directly: $s \circ 1_a = s = s \circ g_a$, so the two would-be distinct factorizations $s = t \circ 1$ and $s = t \circ g$ (with $t = s$ the only candidate transversal) collapse — the right $\mathbb{Z}/2$ -action on the singleton $\{s\}$ is the trivial action, which is not free. Target orientation is symmetric: $1_b \circ s = s = g_b \circ s$, the left action is trivial, and again $2 \nmid 1$. So both orientations fail at once. $\square$

Remark 10 (The one-sided minimal case already breaks). Even the asymmetric category with $\text{End}(a) = \mathbb{Z}/2$, $\text{End}(b)$ trivial, and a single arrow $a \rightarrow b$ fixed by the source loop already fails source-UF: it is the smallest non-poset extension of the walking arrow $(\bullet \rightarrow \bullet)$, and one nontrivial loop acting trivially on a singleton hom-set is enough to destroy uniqueness.

5 Two further nuances

Remark 11 (Unique factorization does not give a poset skeleton). Even when unique factorization holds, the transversal is a genuine *choice* and need not be the reachability poset — it can be *thick*. Let $\text{End}(a)$ be trivial, $\text{End}(b) = \mathbb{Z}/3$, and $\text{Hom}(a, b)$ a set of three arrows on which $\mathbb{Z}/3$ acts *freely* on the left (a single regular orbit). Since $\text{End}(a)$ is trivial, source-UF holds trivially. But the source-transversal is the entire $\text{Hom}(a, b)$: three parallel arrows $a \rightarrow b$, manifestly *not* thin. So clause (c) of the naive conjecture — “skeleton = poset reflection” — fails independently of clauses (a),(b).

Remark 12 (Source and target loops are independent). Source loops act on the right and target loops on the left, and these are logically independent: in general one cannot absorb both into a single one-sided factorization. The naive conjecture conflates the two orientations, whereas $\text{Hom}(a, b)$ genuinely carries *two* actions and freeness on one side says nothing about the other. (In Example 8 both fail; in Remark 11 the right side is trivially free while the left is free but with a thick transversal.)

Remark 13 (Computational evidence). A brute-force script `two_atoms_check.py` enumerates roughly 26 small categories and tests source-oriented unique factorization on each. Of these, 18/26 satisfy source-UF and 8/26 fail; every one of the 8 failures involves a nontrivial loop acting non-freely on a hom-set too small to carry a free orbit — precisely the mechanism of Example 8 and Corollary 7. This matches Theorem 5 exactly: failure $\iff$ non-free connecting biset.

6 The correct general statement

The naive picture is salvaged not as a unique factorization but as a bimodule/profunctor assembly.

Theorem 14 (Corrected decomposition). *Let $\mathcal{C}$ be a small category with object set $\text{ob } \mathcal{C}$.*

1. (General assembly.) $\mathcal{C}$ is reconstructed from the data

$$\left(\text{End}(a) \right)_{a \in \text{ob } \mathcal{C}}, \quad \left(\text{Hom}(a, b) \text{ as an } (\text{End}(b), \text{End}(a))\text{-biset} \right)_{a, b},$$

together with the composition maps $\text{Hom}(b, c) \otimes_{\text{End}(b)} \text{Hom}(a, b) \rightarrow \text{Hom}(a, c)$, as the collage (category of elements / profunctor glueing) of the vertex endomorphism monoids along these connecting bisets. This is a bimodule-style assembly, not a unique factorization, and it always exists.

2. (Zappa–Szép special case.) $\mathcal{C}$ is a Zappa–Szép product of a transversal skeleton and its vertex endomorphism monoids, in the sense of the source-oriented Conjecture 4, if and only if every connecting biset $\text{Hom}(a, b)$ is free as a right $\text{End}(a)$ -set (the one-sided, regular case). Dually for the target orientation and left freeness.

Proof. (1) Composition in $\mathcal{C}$ restricted to $\text{End}(a)$ recovers each vertex monoid, and the biset structure of Definition 2 records pre/post-composition; the full composition maps factor through the relative tensor product over $\text{End}(b)$ because $(g \circ f)$ depends on f, g only up to inserting/removing a b -loop, which is exactly the coequalizer defining $\otimes_{\text{End}(b)}$. Conversely this data plus the composition maps reassembles $\mathcal{C}$: objects are the indices, morphisms $a \rightarrow b$ are the biset elements, identities are the monoid units, and the composition maps give associative composition (associativity is the compatibility of the tensor glueing). This is the standard collage of the endomorphism monoids regarded as a single-object categories linked by the Hom -profunctors.

(2) Apply Theorem 5 at every pair a, b : source-oriented unique factorization globally is exactly the statement that every connecting biset is a free right $\text{End}(\text{source})$ -set, in which case picking a transversal per orbit gives the (thin-or-thick) skeleton and the loops, with composition as the mutual action — a Zappa–Szép product. If some biset is non-free, the factorization fails there by Theorem 5, so no global Zappa–Szép decomposition exists. $\square$

So the two-atoms dream is *true exactly in the free, regular case* and false in general; the universal truth is the weaker bimodule collage.

7 Grant relevance

This converts SEED open-question Q2 — the distributive-law / Zappa–Szép decision procedure for directed containers — into a *checkable criterion*. A Zappa–Szép (equivalently, distributive-law) decomposition of a directed container into a skeleton and its vertex monoids exists if and only if each hom-set is free over the relevant vertex endomorphism monoid. Freeness is finitely checkable (compute orbits, test that each orbit map is injective), and the counting condition $|\text{End}(a)| \mid |\text{Hom}(a, b)|$ of Corollary 7 is a quick necessary pre-test that rejects most non-decomposable cases instantly. For the grant this is a clean, implementable algorithm with a sharp theorem behind it: the decision procedure is exactly an orbit-freeness check on the connecting bisets, and when it fails, the corrected decomposition theorem (Theorem 14) still gives a canonical profunctor-collage assembly to fall back on.